Fusing Narrative Semantics for Financial Volatility Forecasting

Multi-Modal Deep Learning for Dynamic Markets

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Reference: Yaxuan Kong, Yoontae Hwang, Marcus Kaiser, Chris Vryonides, Roel Oomen, and Stefan Zohren. 2025. Fusing Narrative Semantics for Financial Volatility Forecasting. In 6th ACM International Conference on AI in Finance (ICAIF '25), November 15 to 18, 2025, Singapore, Singapore. ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3768292.3771256

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Overview

45 Minutes

01 **The Landscape** — 3 min
LLMs Meet Quantitative Finance

02 **The Challenge** — 5 min
Why Volatility Forecasting Matters

03 **Our Framework** — 15 min
M2VN: Architecture & Design

04 **Experiments** — 8 min
Results & Ablation Studies

05 **The Bigger Picture** — 8 min
Bias & Evaluation Integrity

06 **Looking Forward** — 4 min
Open Problems & Community

LLMs Are Reshaping Quantitative Finance

594%

Growth in LLM-for-Finance papers (2023–2025)

36 → 250 papers at major venues

52%

Financial services firms using generative AI (2024 NVIDIA Survey)

Up from 40% in 2023

6+

Application domains covered by financial LLMs

Sentiment, Time Series Foundation Models, Reasoning, Agents…

- *Kong et al. (2024). A Survey of Large Language Models for Financial Applications. Journal of Portfolio Management (Papers Published into Two Parts in Journal of Portfolio Management).*
- *Kong et al. (2024). Large Language Models for Financial and Investment Management: Applications and Benchmarks (Journal of Portfolio Management).*
- *Kong et al. (2024). Large Language Models for Financial and Investment Management: Models, Opportunities, and Challenges (Journal of Portfolio Management).*
- *Kong et al. (2026). Evaluating LLMs in Finance Requires Explicit Bias Consideration. https://arxiv.org/pdf/2602.14233.*

A Holistic View of Financial LLM Applications

Linguistic Tasks	Sentiment Analysis	Time Series	Reasoning	Agent Modeling
Summarization, NER, classification of financial documents	News, social media, earnings calls, FOMC minutes	Forecasting, anomaly detection, imputation, pattern classification	Planning, recommendation, decision support, real-time Q&A	Trading agents, market simulation, multi-agent systems

Today's Focus: Financial Time Series Analysis

How can we fuse unstructured news narratives with structured price data to improve volatility forecasting — while maintaining temporal integrity?

Why Volatility Forecasting Matters

Critical for Financial Decision-Making

- Risk Management
  - VaR, expected shortfall, portfolio hedging
- Derivative Pricing
  - Options, structured products valuation
- Portfolio Optimization
  - Dynamic allocation under uncertainty
- Regulatory Compliance
  - Capital adequacy, stress testing

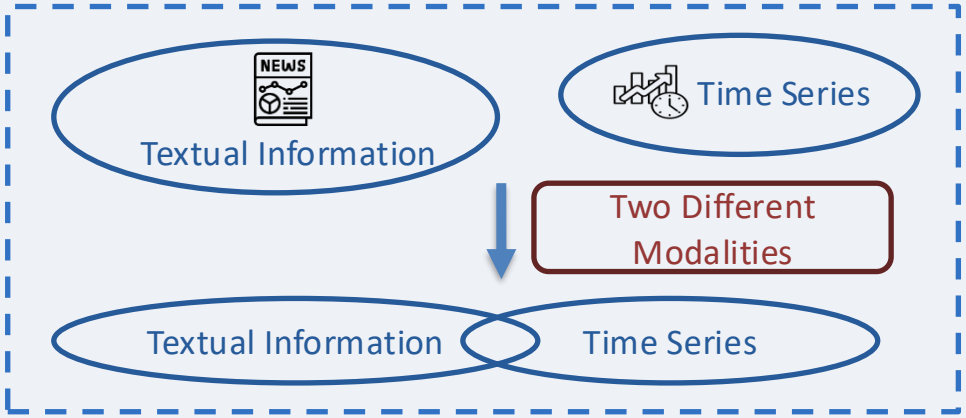
Evolution of Volatility Models

GARCH (1986)	Daily squared returns, linear structure
HAR (2009)	Multi-frequency realized volatility
HAR-X	Exogenous variables: VIX, macro indicators
Deep Learning	Informer, Autoformer, TimesNet, TimeXer…
M2VN (Ours)	Multi-modal: price + news + temporal markers

Two Fundamental Challenges

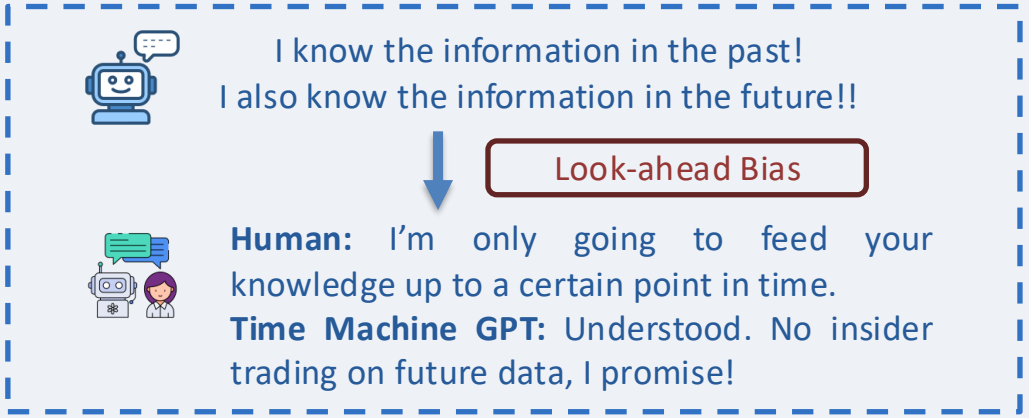
Challenge 1: Modality Fusion

- Financial news and time series are fundamentally different data types
- Text: high-dimensional semantic embeddings
- Price data: low-dimensional numerical features
- Naive concatenation loses complementary signals
- **Need: Learned alignment between modalities**



Challenge 2: Look-Ahead Bias

- Modern LLMs trained on future data (post-cutoff leakage)
- GPT-4 embeddings may encode future events
- RAG systems retrieve documents edited after publication
- This silently inflates backtest performance
- **Need: Point-in-time language model (TiMaGPT)**



Time Machine GPT: Point-in-Time LLMs

Ensuring Temporal Integrity in Financial NLP

How TiMaGPT Works

- Series of LLMs with strict temporal cutoffs
- Each model's training data ends 1 year before prediction date
- Knowledge cutoff advanced year-by-year
- Weights frozen after training — no future leakage
- Open-source: verifiable temporal boundaries

Why This Matters

- Standard LLMs (GPT-4, etc.) trained on data up to recent cutoff
- When backtesting on 2021 data, GPT-4 already "knows" what happened
- This is time travel — model has memorized future events
- TiMaGPT prevents this: for day τ, uses weights that only saw data before τ
- First application of TiMaGPT to volatility forecasting

- *Drinkall et al. (2024). Time Machine GPT. arXiv:2404.18543*

OUR FRAMEWORK

M2VN: Multi-Modal Volatility Network

End-to-end deep learning framework unifying time series features with unstructured news data

Input Data

- Price Data (OHLCV)
- RV (daily, weekly, monthly)
- Momentum indicators
- VIX, News count
- News articles (text)
- Calendar markers

→

Feature Encoder

← Price encoder
(affine norm + spectral)
← News encoder (TiMaGPT)
(projection layer)
← Time encoder
(harmonic embedding)

→

Latent Dynamics

L stacked blocks:
Spectral filtering
(DFT → top-k modes)
Inception-style 2D conv
Residual connections
Gated cross-modal fusion

→

Prediction Head

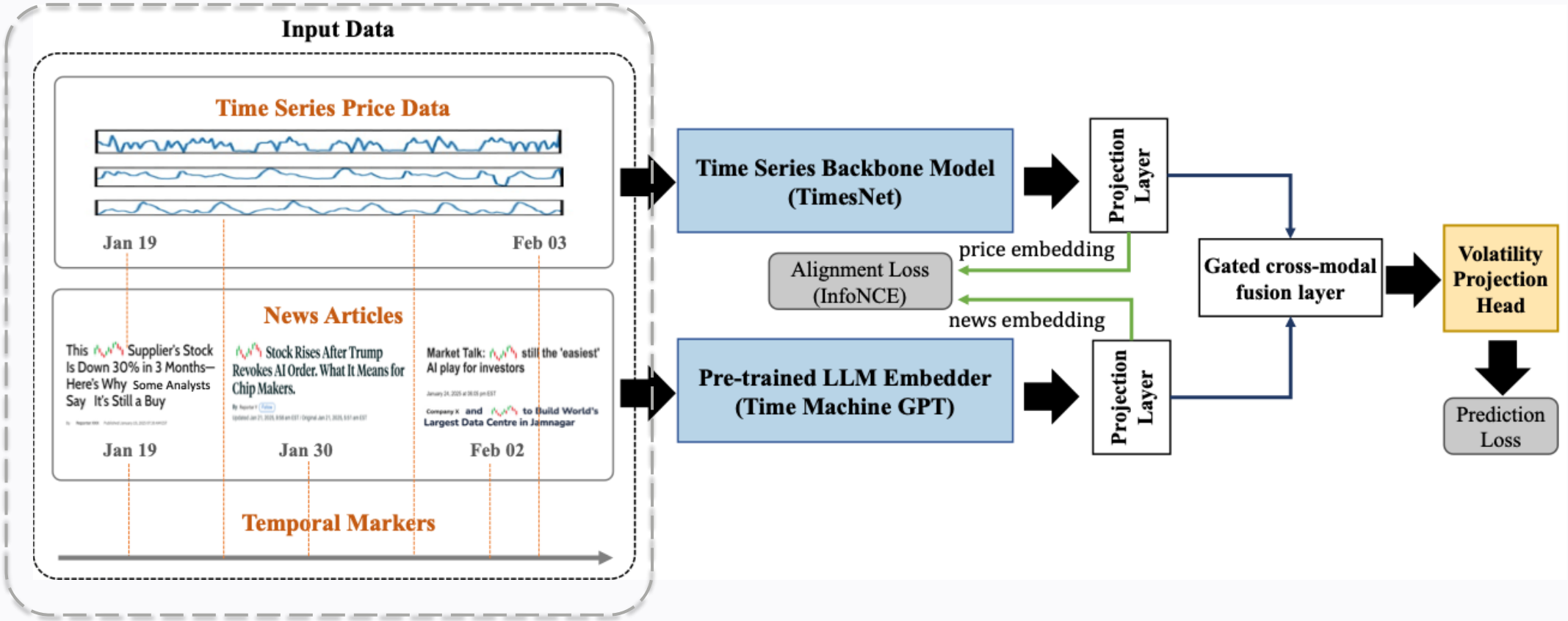
Temporal projection
Extrapolate to horizon H
Affine map to forecast
Output: $\hat{y}(t+H)$
Deterministic point est.

Alignment Loss (InfoNCE) between encoders

Prediction Loss

M2VN: Multi-Modal Volatility Network

End-to-end deep learning framework unifying time series features with unstructured news data



OUR FRAMEWORK

Multi-Modal Feature Encoder

Price Encoder

E_price

- Input: $X \in R^{(T \times d_x)}$
- Affine normalization
- Periodic-spectral operator
- Decomposes trajectory into dominant frequency channels
- Output: $h^{price} \in R^{(T \times d)}$

News Encoder

E_news^LLM

- Input: $N \in R^{(T \times d_n)}$
- TiMaGPT embeddings per day
- Titles + bodies of articles
- Projection layer
- Captures semantic drift
- Output: $h^{news} \in R^{(T \times d)}$

Time Encoder

E_time

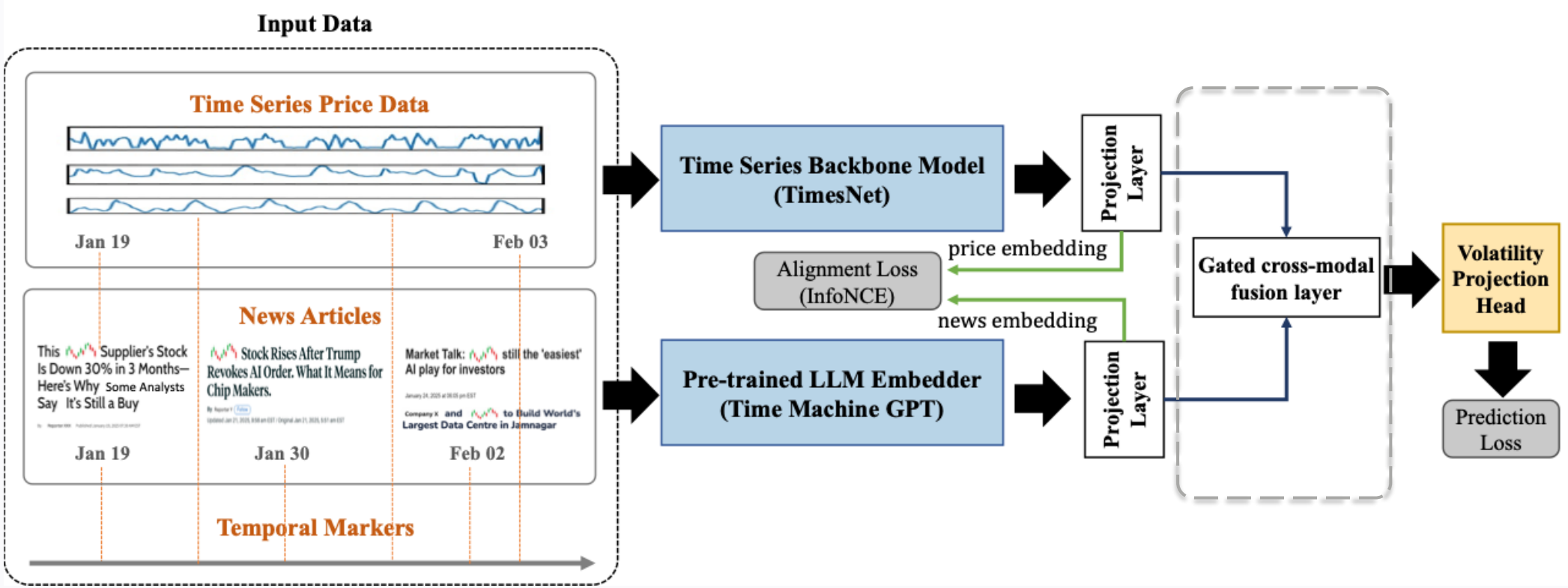
- Input: $M \in R^{(T \times d_m)}$
- Calendar features: Day of week, month, year
- Harmonic embedding
- Captures market seasonality
- Output: $h^{time} \in R^{(T \times d)}$

```
h(1:T) = [h_price; h_news; h_time]   →   Raw input for temporal reasoning
```

OUR FRAMEWORK

M2VN: Multi-Modal Volatility Network

End-to-end deep learning framework unifying time series features with unstructured news data



Latent Dynamics Module

Step 1: Spectral Filtering

- Transform latent trajectory to Fourier domain via DFT
- Keep only top-k modes with highest energy
- Provably optimal k-term approximation (Proposition 3.1)
- Built-in denoising: separates signal from stochastic noise

Step 2: Inception-Style 2D Convolution

- Multi-scale kernels across the period-channel grid
- Captures residual cross-temporal interactions
- Isolates k most energetic periodic components
- Causal truncation: no future information leaks

Step 3: Gated Cross-Modal Fusion

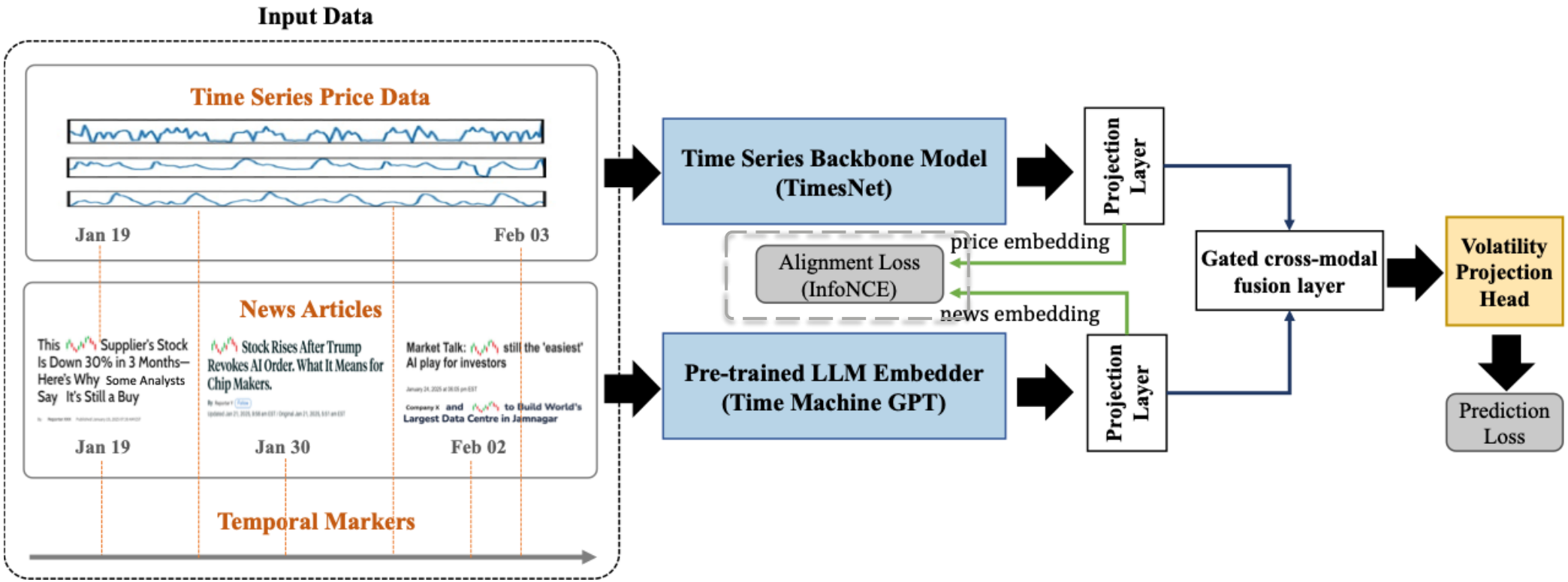
$$a(1:T) = \sigma(W_g[r(1:T); t(1:T)]) \in (0,1)^{(T\times 1)}$$

$$z(1:T) = a \odot r + (1-a) \odot t$$

- Dynamically weights price vs. news representations
- Combined with bilinear and absolute-difference interactions
- Concatenated and linearly projected back to R^d
- Output: both price-aware and news-aware semantics

M2VN: Multi-Modal Volatility Network

End-to-end deep learning framework unifying time series features with unstructured news data



Auxiliary Alignment Loss

InfoNCE Contrastive Objective

- Intuition: At the same timestep τ, the price representation and news representation should be similar, while representations from different timesteps should be dissimilar.

$$\mathcal{L}_{\text{align}}(\theta) = \mathbb{E}[-\log \frac{\exp(\langle \mathbf{r}_\tau, \mathbf{t}_\tau \rangle / \tau)}{\sum_{\tau'} \exp(\langle \mathbf{r}_\tau, \mathbf{t}_{\tau'} \rangle / \tau)}],$$

τ > 0 is a learned temperature parameter

Dual-Objective Training

- Total loss:

$$J(\theta) = L_pred(\theta) + \lambda \cdot L_align(\theta)$$

- L_pred: Mean Squared Error on volatility
- L_align: InfoNCE cross-modal alignment
- λ: Hyperparameter controlling alignment weight
- Multi-task paradigm: predict + align

Why Does Alignment Matter?

- Without alignment: model learns modality-specific shortcuts rather than joint representations
- Ablation shows removing alignment loss degrades QLike by 11.7% and MAPE by 3.6% on average
- Forces discovery of complex, non-linear interactions between market dynamics and news

Experimental Setup

Data

- 7 large-cap U.S. equities: KO, CMCSA, COP, GILD, MRK, NKE, ORCL
- Daily OHLCV + full news articles from FNSPID dataset
- Train: Jan 2013 – Dec 2017 | Val: 2018 – 2020 | Test: 2021 – 2023
- All news temporally aligned: only articles published on/before prediction day

Baselines (10 Models)

- Classical: HAR, HAR-X (OLS), HAR-X (Ridge), HAR-X (Lasso)
- Deep Learning: Informer, Autoformer, DLinear, TimesNet, PAttn, TimeXer
- HAR-X baselines use PCA-reduced news embeddings
- All DL baselines ingest raw TiMaGPT embeddings directly

Evaluation Metrics

- MAPE: Scale-free percentage error (symmetric)
- QLike: Quasi-likelihood loss (penalizes under-prediction of high-vol episodes)

Training Details

- Optimizer: Adam (lr = 3×10^{-4}, cosine decay)
- Sequence length: 12-time steps | 100-config random search | 3 seeds

Main Results: M2VN vs. 10 Baselines

71.4%
Top rank across all settings

12/14
Best or 2nd-best (7 stocks × 2 metrics)

↓ 2.86%
Avg MAPE reduction vs. ablated model

↓ 5.64%
Avg QLike reduction vs. ablated model

Selected Results: QLike (lower is better)

Method	KO	CMCSA	COP	GILD	MRK	NKE	ORCL
HAR-X (Lasso)	0.0627	0.0692	0.0594	0.0716	0.0685	0.0646	0.0809
TimesNet	0.0666	0.0833	0.0652	0.0660	0.0760	0.0679	0.0864
TimeXer	0.0661	0.0807	0.0589	0.0673	0.0797	0.0671	0.0849
M2VN (Ours)	**0.0589**	**0.0769**	**0.0552**	**0.0630**	**0.0679**	**0.0594**	**0.0696**

Ablation Studies: What Matters?

Is Volume a Proxy for News?

Answer: No. Volume and news are complementary.

- Removing volume degrades 27/28 stock-metric pairs
- Average MAPE reduction with volume: 2.86 ± 3.06%
- Average QLike reduction with volume: 5.64 ± 2.40%
- Volume captures: liquidity, disagreement, market attention
- News captures: semantic shifts, event-driven sentiment

Both are essential, non-redundant signals

Is News Embedding Predictive?

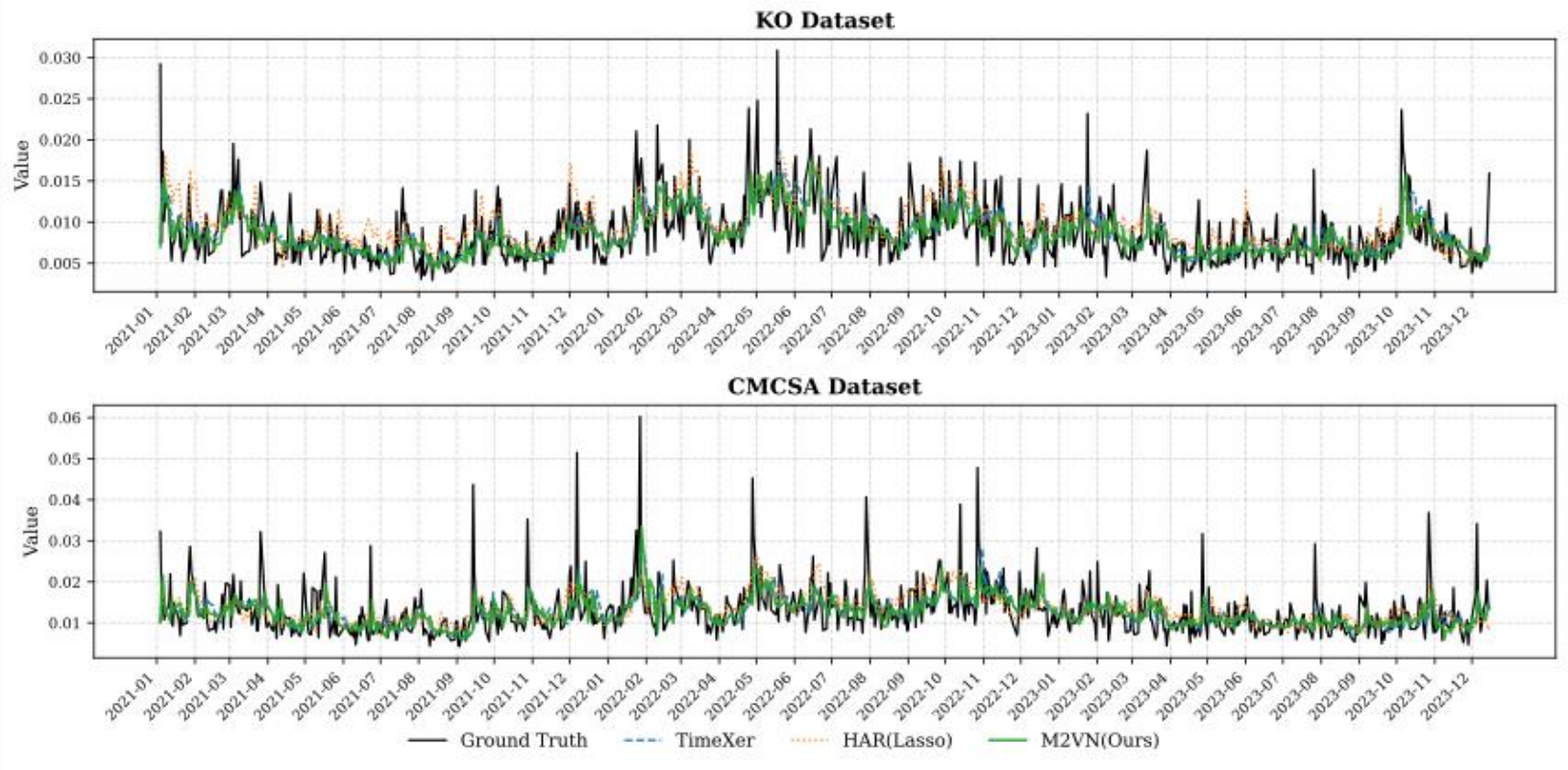
Answer: Yes. Universal improvement across all stocks.

- KO: QLike 0.0641 → 0.0589 (news added)
- COP: QLike 0.0599 → 0.0552 (news added)
- ORCL: QLike 0.0780 → 0.0696 (news added)
- Stocks with high news sensitivity (COP, ORCL) benefit most
- GILD shows smaller gains → less news-driven volatility

News provides qualitative insights not in price data alone

Forecast Quality: Tracking Volatility Shocks

Visual comparison on KO and CMCSA datasets (2021–2023 out-of-sample)



Sharp Peaks

M2VN captures the direction and timing of volatility spikes (e.g., CMCSA spike in early 2022) while baselines produce smoothed, delayed responses.

Calm Periods

During low-volatility regimes, M2VN avoids the persistent over-prediction seen in HAR-X (Lasso) and the delayed responses of TimeXer.

Extreme Events

All models struggle with exact magnitudes of extreme single-day shocks, but M2VN most reliably identifies the onset and direction of sharp movements.

But Can We Trust These Results?

Stepping back from M2VN to ask a broader question about the entire field of LLM-for-Finance research…

Evaluating LLMs in Finance Requires Explicit Bias Consideration

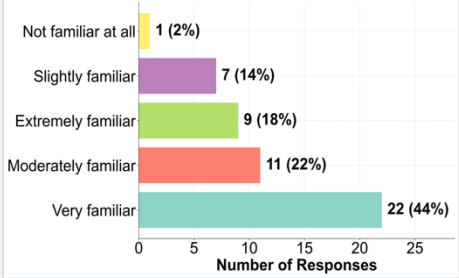
Kong, Lee, Hwang, et al. (2025). https://arxiv.org/pdf/2602.14233

The Five "Sins" of Financial LLM Evaluation

1

Look-Ahead Bias

Model uses future information via weights or retrieved context

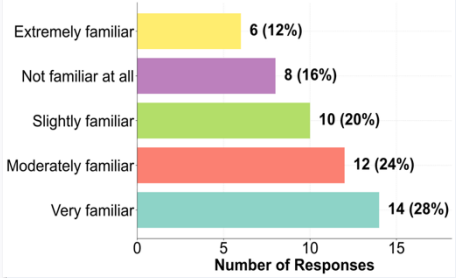


(survey, n=50)

2

Survivorship Bias

Evaluation excludes delisted/failed firms from the universe

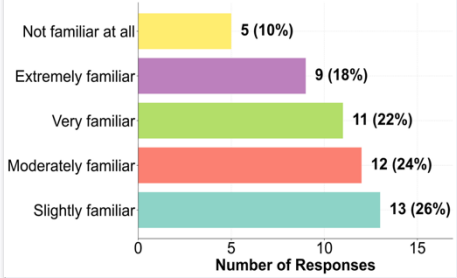


(survey, n=50)

3

Narrative Bias

LLM generates plausible but ungrounded causal explanations

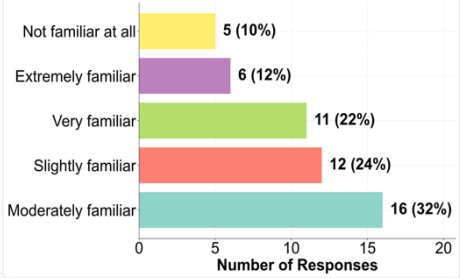


(survey, n=50)

4

Objective Bias

Training rewards confidence over calibrated uncertainty

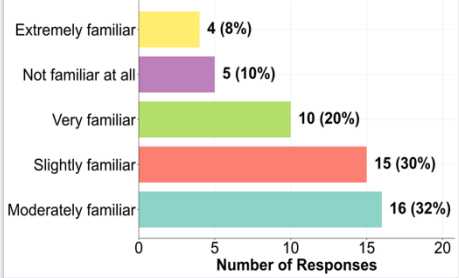


(survey, n=50)

5

Cost Bias

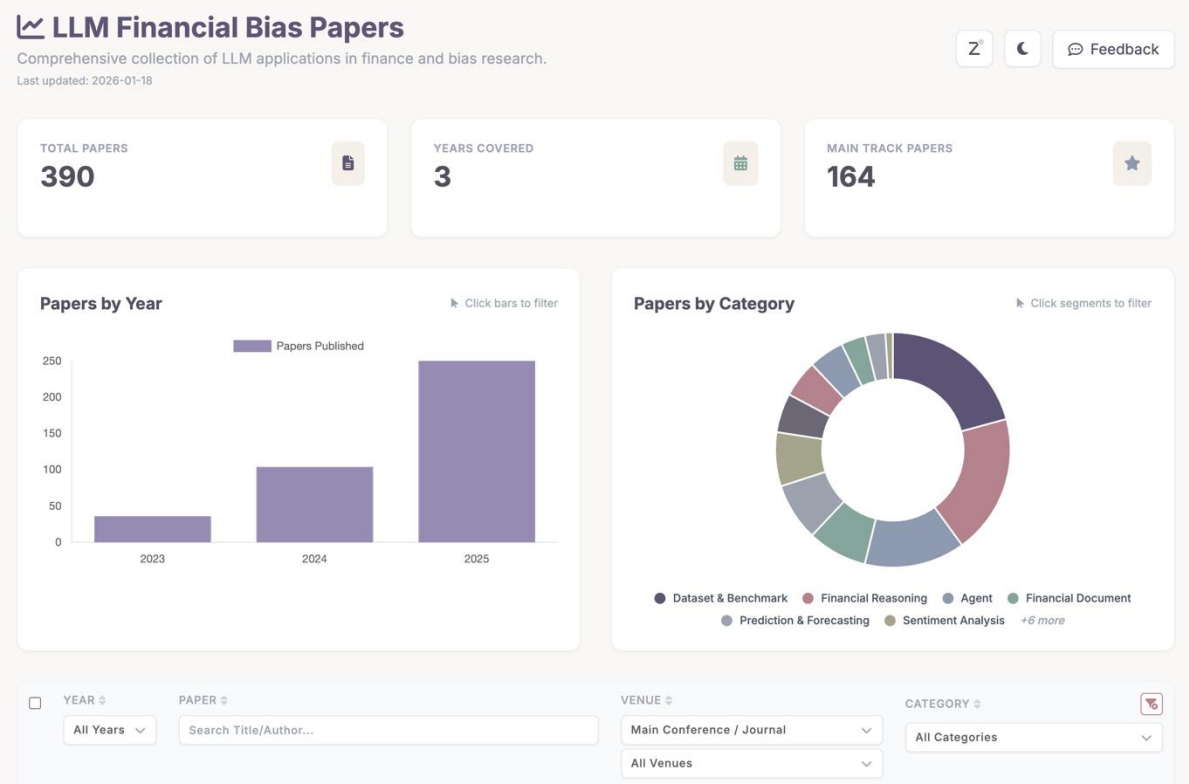
Reports ignore latency, inference cost, and trading friction



(survey, n=50)

164 papers reviewed (2023–2025): No single bias discussed in more than 28% of studies.
These biases compound to create an illusion of validity.

Research Literature (LLM in Finance)



2025	**Efficient Multi-Expert Tabular Language Model for Banking** Yue Guo, Wentao Zhang, Xiaojun Zhang, Vincent W. Zheng, Yi Yang	KDD (ADS Track) - Aug 2024 - cycle 1	Financial Reasoning
2025	**FinBERT2: A Specialized Bidirectional Encoder for Bridging the Gap in Finance-Specific Deployment of Large Language Models** Xuan Xu; Fufang Wen; Beilin Chu; Zhibing Fu; Qinhong Lin; Jiaqi Liu; Binjie Fei; Yu Li; Linna Zhou; Zhongliang Yang.	KDD (ADS Track) - Feb 2025 - cycle 2	Financial Reasoning
2025	**FLAG: Fraud Detection with LLM-enhanced Graph Neural Network** Chengdong Yang; Hongrui Liu; Daixin Wang; Zhiqiang Zhang; Cheng Yang; Chuan Shi.	KDD (ADS Track) - Feb 2025 - cycle 2	Risk Management
2025	**AlphaAgent: LLM-Driven Alpha Mining with Regularized Exploration to Counteract Alpha Decay** Ziyi Tang, Zechuan Chen, Jiarui Yang, Jiayao Mai, Yongsen Zheng, Keze Wang, Jinrui Chen, Liang Lin	KDD (Research Track) - Feb 2025 - cycle 2	Agent
2025	**CAMEF: Causal-Augmented Multi-Modality Event-Driven Financial Forecasting by Integrating Time Series Patterns and Salient Macroeconomic Announcements** Yang Zhang, Wenbo Yang, Jun Wang, Qiang Ma, Jie Xiong	KDD (Research Track) - Feb 2025 - cycle 2	Prediction & Forecasting
2025	**VideoConviction: A Multimodal Benchmark for Human Conviction and Stock Market Recommendations** Michael Galarnyk, Veer Kejriwal, Agam Shah, Yash Bhardwaj, Nicholas Watney Meyer, Anand Krishnan, Sudheer Chava	KDD (Datasets & Benchmarks) - Feb 2025 - Cycle 2	Dataset & Benchmark

Showing 1 to 20 of 164 results

Prev 1 2 3 4 5 ... 9 Next

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The paper's date, link, and venue have been verified by human reviewers; however, errors may still occur. Feedback is welcome.

164 papers reviewed (2023–2025): No single bias discussed in more than 28% of studies.
These biases compound to create an illusion of validity.

The Structural Validity Framework

A checklist for interpreting backtest performance as evidence of deployment readiness

Temporal Sanitation	Dynamic Universe	Rationale Robustness	Epistemic Calibration	Realistic Constraints
Non-Anticipativity	*Survivorship Control*	*Causal Validity*	*Uncertainty*	*Cost & Latency*
Prevent future info via weights or retrieval.	Time-indexed tradable universe.	Ground claims in evidence.	Output distributions, not just point estimates.	Include transaction costs.
Enforce knowledge cutoffs.	Include delisted firms.	Require audit trails.	Allow abstention.	Report inference latency.
Use point-in-time data.	Sample w/o proxy for survival.	Test entity substitution.	Score refusal fairly.	Budget-matched baselines.
		Detect narrative bias.		Measure net utility.

M2VN addresses checks 1 & 2 via TiMaGPT (temporal sanitation) and temporal alignment (no survivorship bias in news).

The Structural Validity Framework

Financial LLM Structural Validity Checklist

Self-Assessment Tool for Rigorous Financial Decision-Making Systems

Validity Score: 64% (7/11)

Reset Form | Download PDF

Strict Adherence Required: This framework sets minimum requirements for interpreting a backtest as evidence of deployment performance.

1. Temporal Sanitation — Non-Anticipativity

Requirement	Rationale	Assessment
Parametric Knowledge Cutoff Disclosure Are the latest dates of pre-training and fine-tuning disclosed? Does the evaluation window strictly follow these dates?	Sin #1: Look-Ahead Bias Prevents "Time Travel" where the model uses future knowledge embedded in its weights to predict the past.	Select Status... Notes (e.g., cutoff dates)...
Point-in-Time (PIT) Retrieval Are retrieval sources constructed using ONLY documents available at the specific decision time? (No live web search or mutable wikis).	Sin #1: Look-Ahead Bias Prevents "External Leakage" where search engines rank historical docs based on future popularity or edits.	Pass Notes on data sources...
Full Trace Logging Are prompts, retrieved documents, tool outputs, and timestamps logged for temporal auditing?	Sin #1: Look-Ahead Bias Ensures verifiability of the information set available at every specific decision step.	Pass Notes...

2. Dynamic Universe Construction — Survivorship Control

Time-Indexed Tradable Universe Does the simulation sample from the universe active at the decision time, including delisted or failed entities?	Sin #2: Survivorship Bias Ensures the model faces the true probability of failure inherent in financial markets.	Fail Notes on universe handling...
Unbiased Task Sampling Are benchmarks constructed without conditioning on ex-post news volume or present-day prominence?	Sin #2: Survivorship Bias Prevents testing only on "winners" that generated enough historical news to be remembered.	Pass Notes...
Failure Regime Reporting Does the study report outcomes separately for surviving vs. non-surviving entities?	Sin #2: Survivorship Bias Prevents inflation of performance metrics by hiding losses from failed firms.	N/A We used WRDS dataset.

3. Rationale Robustness — Causal Validity

Evidence Grounding & Auditing Are rationales explicitly grounded in retrieved documents? Are claims audited for factual hallucinations?	Sin #3: Narrative Bias Prevents the "Illusion of Understanding" where models suture disparate signals into false but coherent narratives.	Pass Notes...
Negative Controls & Entity Substitution Does the system refrain from generating detailed causal stories when inputs are scrambled or entities are substituted?	Sin #3: Narrative Bias Tests if the "reasoning" is genuine or just a retrieval of memorized company-specific text patterns.	Pass Notes...

4. Epistemic Calibration — Uncertainty & Abstention

Explicit Abstention Option Does the action space include "No Trade" or "Do Not Know", and is it scored as a legitimate outcome?	Sin #4: Objective Bias Prevents the model from being forced to hallucinate or guess to satisfy a "helpful assistant" objective.	Pass Notes...
Calibrated Confidence/Distribution Does the system output a predictive distribution or a structured uncertainty estimate?	Sin #4: Objective Bias Financial decisions require calibrated probability, not just fluent text (verbalized overconfidence).	Pass Notes...

5. Realistic Implementation — Cost & Latency

Latency-Aware Execution Are trades executed at the price available AFTER generation is complete (accounting for slippage), not the price at the start?	Sin #5: Cost Bias (Latency) Zero-latency assumptions overestimate returns for time-sensitive strategies given the slow inference of LLMs.	Fail Notes...
Net Utility Reporting Are metrics reported strictly after deducting transaction costs and operational inference costs?	Sin #5: Cost Bias (Monetary) Prevents complex, expensive pipelines from appearing superior to simple baselines when they are economically unviable.	Fail Notes...

What Practitioners Are Saying

Survey of 50 researchers and practitioners working with financial LLMs

76%

say evaluation tools are scarce or non-existent

50%

identify "lack of frameworks" as the biggest bottleneck

62%

have low confidence that benchmarks capture reality

48%

say current research gives little bias guidance

The Gap: Tools vs. Awareness

- Academics recognize biases by name but struggle to diagnose them systematically
- Industry practitioners see the consequences but lack standardized evaluation protocols
- The community needs shared, open-source frameworks for rigorous financial LLM evaluation
- This is precisely the kind of challenge that benefits from open, collaborative research efforts

LOOKING FORWARD

Open Problems & Future Directions

Multi-Asset Extension

- Extend M2VN beyond single-stock volatility to cross-asset and portfolio-level forecasting.
- Capture contagion and spillover effects.

Richer Modalities

- Incorporate earnings calls (audio), SEC filings, social media sentiment.
- Explore vision models for chart pattern recognition.

Uncertainty Quantification

- Move from point estimates to predictive distributions.
- Essential for downstream risk management applications.

Evaluation Standards

- Adopt the Structural Validity Checklist as community practice.
- Build shared benchmarks with temporal integrity baked in.

We believe these challenges are best addressed through open research communities where practitioners share unsolved problems and researchers contribute solutions. We are building exactly such a community — one that delivers practical solutions to both new and longstanding questions.

VulpiVox Lab → vulpivox.com

Key Takeaways

1	**Multi-modal fusion works**	M2VN outperforms 10 baselines across 7 stocks by combining time series with LLM news embeddings via gated fusion and contrastive alignment.
2	**Temporal integrity is non-negotiable**	Using TiMaGPT (point-in-time LLM) prevents look-ahead bias — without this, any backtest result is suspect.
3	**Volume and news are complementary**	Trading volume is not a proxy for news. Both modalities contribute unique, non-redundant predictive signals for volatility.
4	**The field needs evaluation standards**	Five recurring biases undermine financial LLM research. The Structural Validity Framework provides concrete, actionable diagnostics.
5	**Open collaboration accelerates progress**	The gap between tools and awareness demands shared frameworks, open-source benchmarks, and community-driven research.

Thank You

Questions & Discussion

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Papers & Resources

M2VN: https://github.com/Eleanorkong/M2VN-Multi-Modal-Learning-Network-for-Volatility-Forecasting

Bias Framework: https://github.com/Eleanorkong/Awesome-Financial-LLM-Bias-Mitigation

LLM-Finance Survey: Large Language Models for Financial and Investment Management: Applications and Benchmarks (https://www.pm-research.com/content/iijpormgmt/51/2/162). Large Language Models for Financial and Investment Management: Models, Opportunities , and Challenges. (https://web.media.mit.edu/~xdong/paper/jpm24c.pdf).

Open Research Community: vulpivox.com

FAQ

How Do News Embeddings Enter the Model?

Q: You use LLM embeddings instead of a scalar sentiment score — how exactly do they flow through the architecture?

1 Text → Embedding

- Each news article (title + body) is fed through TiMaGPT — a point-in-time LLM whose training data ends one year before the article date.
- Token-level outputs are mean-pooled into a single 768-dim vector per article.
- If a stock has 5 articles on a given day, those 5 vectors are averaged into one daily title embedding and one daily body embedding.

2 Dual-Track Encoding

- Title (768d) and body (768d) are concatenated → 1,536 dims, then linearly projected to match the price branch's latent size.
- The price branch independently encodes OHLCV + momentum + VIX through a TimesNet backbone (FFT → 2D convolution).
- Both branches output a representation per timestep.

3 Fusion: Align + Gate + Inject

- InfoNCE (Information Noise-Contrastive Estimation) loss pulls same-day price and news representations together, pushes different-day pairs apart (news are shifted by one day).
- A learned gate decides how much to trust price vs. news at each timestep. Element-wise product and absolute difference capture agreement with divergence.
- The fused result is added as a residual on top of the price branch — if news adds nothing, the gate can zero it out.

FAQ

What Happens When News Is Sparse?

Q: Some days have no news articles for a given stock. How does M2VN handle missing or sparse news?

How Sparse Days Are Handled

- Article_count (C_tau) is an explicit input feature, so the model knows when news is sparse
- On zero-article days, the embedding is a zero vector (or previous day carried forward)
- The gated fusion mechanism learns to down-weight the news channel when it carries no signal
- Gate value a_tau near 1.0 means 'trust price only'; near 0.0 means 'trust news more'

Empirical Evidence

- GILD has lower news volume than COP or ORCL, yet M2VN still outperforms baselines on GILD
- The improvement is smaller for GILD (less news-sensitive), confirming the gate adapts to news availability
- The model degrades gracefully: it never performs worse than the price-only ablation, even on low-news stocks
- This is by design: the gated residual connection means news is additive, not required

Design Principle: Graceful Degradation

M2VN is designed so that news is never required for a prediction. The gated fusion adds a residual contribution on top of the price branch. If the news branch carries no useful signal on a given day, the gate can effectively zero it out, and the model falls back to the price-only representation. This means the model is always at least as good as its price-only version, with news providing an incremental boost when available.

FAQ

How Is Temporal Alignment Handled?

Q: News arrives irregularly. How do you align it with daily price data — and prevent look-ahead bias?

Daily Alignment by Aggregation

- No carry-forward or decay weighting needed — the alignment is structural.
- All news articles published on day t are embedded individually by TiMaGPT, then mean-pooled into a single daily vector.
- This gives one fixed-size embedding per trading day, perfectly aligned with that day's OHLCV row.
- Article_count is included as a separate feature, so the model knows whether a day had 1 article or 20 — capturing information density without inflating the embedding.
- News shifted by one day to account for look-ahead bias.

Three Layers Against Look-Ahead

① LLM Layer

- 2015 news is processed by TiMaGPT-2014 — a model whose training ended a year earlier. The LLM itself has never seen 2015 events.

② Feature Layer

- Price features are pre-lagged: RV_lag1, RV_lag5, RV_lag22. What sits in day t's row is always information available at t.

③ Window Layer

- Input = day [t−12 … t−1], target = day t. Day t's news never appears in the input that predicts day t's volatility — the sliding window naturally excludes it.

FAQ

Does M2VN Produce Distributional Forecasts?

Q: Does M2VN output a full predictive distribution? How does the news channel perform during tail-risk episodes?

Current Design: Point Forecasts

- M2VN outputs a single scalar: tomorrow's predicted realized volatility.
- The loss is MSE + InfoNCE alignment — no quantile or distributional component.
- Evaluation uses QLike and MAPE, both point-forecast metrics.
- This means we currently cannot report calibration, prediction intervals, or tail-specific performance.

What We Know & What's Next

- Ablation evidence: removing news degrades average QLike by ~8% and MAPE by ~6%. The gated fusion dynamically up-weights news when it's informative.
- Visual inspection shows M2VN tracks volatility spikes (e.g., CMCSA early 2022) better than baselines — suggesting news helps at turning points.
- Distributional extension is an explicit open problem (Slide 25): replace MSE with quantile regression or a parametric distribution head to enable proper calibration and tail-risk analysis.

FAQ

How Is Realized Volatility Calculated?

Q: You predict realized volatility, not implied vol. How is the target constructed, and why not use high-frequency data?

Target: Aggregated Range-Based Estimators

- **Three estimators combined via weighted average:**
  - Parkinson (1980): uses daily High and Low range
  - Garman-Klass (1991): uses Open, High, Low, Close
  - Rogers-Satchell (1991): allows non-zero drift
  - **y(t+H) = f_agg(V^P, V^GK, V^RS)**
- **Aggregation produces a more stable measure than any single estimator**
  - All computed from end-of-day OHLC prices (no tick data needed)
  - Range-based estimators are ~5x more efficient than close-to-close squared returns

Why Not High-Frequency (Tick) Data?

- **Practical considerations:**
  - HF data is expensive, proprietary, and not universally available
  - Molnar & Vyrost (2021): for horizons beyond one month, low-frequency estimators match HF accuracy
- **Our design choice:**
  - Maximize accessibility: OHLCV data is freely available for all tickers
  - Focus research effort on the multi-modal fusion, not data sourcing
  - HF extension is a natural next step for intraday applications
  - Key insight: daily OHLC bars are surprisingly powerful for volatility estimation

Stefan Lyocsa, Peter Molnar, and Tomas Vyrost. 2021. "Stock market volatility forecasting: Do we need high-frequency data?" International Journal of Forecasting, 37(3), 1092 to 1110.

Volatility Estimator	Original Formula
Parkinson Volatility	$P_t = \sqrt{\frac{1}{4\log(2)} \log\left(\frac{h_t}{l_t}\right)^2}$
Garman-Klass Volatility	$GK_t = \sqrt{\frac{1}{2}\log\left(\frac{h_t}{l_t}\right)^2 - (2\log(2)-1)\log\left(\frac{c_t}{o_t}\right)^2}$
Rogers-Satchell Volatility	$RS_t = \sqrt{\log\left(\frac{h_t}{o_t}\right)\log\left(\frac{h_t}{c_t}\right) + \log\left(\frac{l_t}{o_t}\right)\log\left(\frac{l_t}{c_t}\right)}$
RB (Average of above)	$RB_t = \frac{P_t + GK_t + RS_t}{3}$

Availability Public websites offering free access to low-frequency data make the P_t, GK_t and RS_t estimators cost-effective and easy to compute. In this study, all low-frequency data was sourced from Yahoo! Finance. For comparative purposes, we also include the average

$$RB_t = \frac{P_t + GK_t + RS_t}{3}$$

considered by Lyócsa et al. (2021a) and Lyócsa et al. (2021b). As argued by the authors, in practice, this simple combination of low-frequency volatility estimators might improve on its components (individual estimators).

Parkinson volatility estimator: Michael Parkinson. 1980. The extreme value method for estimating the variance of the rate of return. Journal of Business, 53(1), 61 to 65.
Garman-Klass volatility estimator: Mark B Garman and Michael J Klass. 1980. On the estimation of security price volatilities from historical data. Journal of Business, 53(1), 67 to 78.
Rogers-Satchell volatility estimator: L Christopher G Rogers and Stephen E Satchell. 1991. Estimating variance from high, low and closing prices. The Annals of Applied Probability, 1(4), 504 to 512.
Low-frequency vs. high-frequency comparison: Stefan Lyocsa, Peter Molnar, and Tomas Vyrost. 2021. Stock market volatility forecasting: Do we need high-frequency data? International Journal of Forecasting, 37(3), 1092 to 1110.
Aggregation of multiple estimators (f_agg): Adam Clements and Daniel PA Preve. 2021. A practical guide to harnessing the HAR volatility model. Journal of Banking & Finance, 133, 106285.

FAQ

Why These 7 Stocks? Does It Generalize?

Q: You test on 7 U.S. large-caps. How confident can we be that M2VN works beyond this sample?

Why These 7?

- Selected to span sectors: consumer staples (KO), media (CMCSA), energy (COP), biotech (GILD), pharma (MRK), retail (NKE), tech (ORCL)
- All have sufficient daily news volume in FNSPID dataset
- Large-cap ensures deep, liquid markets where price formation is well-studied
- Same tickers used in prior volatility forecasting literature for comparability

Generalization Considerations

- Architecture is ticker-agnostic: no stock-specific parameters
- News encoder (TiMaGPT) handles any English-language text
- Cross-sector consistency (COP vs. GILD) suggests the fusion mechanism adapts to different news-volatility regimes
- Open question: performance on small-caps, non-U.S. equities, and multi-asset portfolios

Honest Limitations

- 7 stocks is a small universe. We cannot claim broad market generalization from this alone.
- All stocks are U.S. large-caps with high analyst coverage. Performance on information-sparse or emerging-market equities is unknown.
- FNSPID news coverage varies by ticker. Tickers with sparse coverage may not benefit as much from the news channel.
- Multi-asset extension (equities + fixed income + commodities) is an explicit open problem.

FAQ

How Sensitive Is M2VN to Hyperparameters?

Q: You search over 100 configurations. How robust are the results to hyperparameter choices?

Hyperparameter Search Grid

- Latent dimension d: {12, 24, 32, 64}
- Alignment dimension d_a: {32, 64, 128, 256}
- Temperature τ: {0.01, 0.03, 0.07}
- Inception hidden width d_i: {24, 32, 64, 128, 256, 512}
- Top-k spectral modes: {4, 5, 6}
- 100 random configurations per stock
- Each repeated with 3 random seeds; mean reported

Evidence for Robustness

- Standard deviations across runs are sufficiently small to omit from the table (paper Section 4.2)
- M2VN ranks #1 in 71.4% of settings, top-2 in 85.7%, across diverse stocks with different volatility dynamics
- Consistent improvement across all 7 stocks suggests the architecture, not the hyperparameters, drives performance
- Ablation results (removing volume, news, alignment) show large, consistent degradation, confirming structural contributions

Critical Design Choices

- Early stopping on the validation set prevents overfitting. Adam optimizer with cosine learning rate decay (initial lr = 3e-4).
- All deep learning baselines undergo equivalent architecture-specific random searches to ensure a fair comparison.
- The alignment weight λ controls the trade-off between prediction accuracy and cross-modal coherence.

FAQ

Why Not Just Use Sentiment Scores?

Q: Why use full 768-dim LLM embeddings instead of a simple positive/negative sentiment score?

Typical Sentiment Approaches

- Extract a scalar sentiment score (+1 / 0 / -1) per article
- Aggregate daily (average or weighted sum)
- Feed as a single exogenous feature into HAR-X or similar
- **What is lost:**
- Topical content (what the article is about)
- Entity relationships (who is affected)
- Narrative framing and emphasis
- Semantic drift over time (how the story evolves)

M2VN: Dense LLM Embeddings

- 768-dimensional dense vector per day from TiMaGPT
- Encodes full semantic content, not just valence
- Projection layer learns which dimensions matter for volatility

What is gained:

- Distinguishes oil prices rose due to demand from oil prices rose due to supply shock (same sentiment, different vol implications)
- Captures gradual narrative shifts that precede regime changes
- The InfoNCE loss ensures embeddings align with price dynamics

Empirical Evidence

- HAR-X baselines in our study already use PCA-reduced TiMaGPT embeddings as exogenous features, yet M2VN consistently outperforms them. The advantage comes from the learned alignment and gated fusion, not merely from having richer input features.

FAQ

What Is TimesNet and Why Use It?

Q: M2VN uses TimesNet as the price branch backbone. What is TimesNet and why was it chosen?

TimesNet: Temporal 2D-Variation Modeling

- Introduced by Wu et al. (ICLR 2023) from Tsinghua University
- Key insight: real-world time series exhibit multi-periodicity (daily, weekly, monthly cycles)
- Decomposes temporal patterns into intraperiod variations (within a cycle) and interperiod variations (across cycles)
- Achieves state-of-the-art across five tasks: short/long-term forecasting, imputation, anomaly detection, classification

How TimesNet Works

- **Step 1: Discover periods via FFT**
  - Apply Fast Fourier Transform to find dominant periodicities (e.g., 5-day weekly, 22-day monthly)
- **Step 2: Reshape 1D to 2D tensors**
  - Fold the 1D time series into 2D grids (rows = interperiod, columns = intraperiod)
- **Step 3: Apply 2D inception convolution**
  - Use parameter-efficient inception blocks with multi-scale 2D kernels to capture patterns

Why TimesNet in M2VN?

- Volatility exhibits strong multi-frequency periodicity: TimesNet's FFT-based period discovery naturally captures daily, weekly, and monthly cycles
- The 2D convolution acts as built-in spectral filtering, which M2VN extends with explicit top-k mode selection (Proposition 3.1)
- TimesNet's modular design allows clean integration with the news encoder through the gated cross-modal fusion layer
- The inception architecture is parameter-efficient, reducing overfitting risk on the relatively small per-stock datasets

Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long. 2023. TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis. In Proceedings of the International Conference on Learning Representations (ICLR 2023).

FAQ

What Is the HAR Model and Its Variants?

Q: You benchmark against HAR and HAR-X. What are these models and why are they strong baselines?

HAR: Heterogeneous Auto-Regressive Model

- Introduced by Corsi (2009), building on the Heterogeneous Market Hypothesis
- Predicts current RV as a linear function of lagged RV at three frequencies:
- $RV(t+1) = c + \beta_d \cdot RV_daily + \beta_w \cdot RV_weekly + \beta_m \cdot RV_monthly$
- Captures the well-known long-memory property of volatility
- Simple OLS estimation, highly interpretable, surprisingly competitive

HAR-X: Extended with Exogenous Variables

- HAR-X adds external predictors beyond lagged RV:
- Implied volatility (VIX), macro indicators, news features

Three estimation variants in our study:

- HAR-X (OLS): standard least squares
- HAR-X (Ridge): L2 regularization, shrinks coefficients
- HAR-X (Lasso): L1 regularization, performs variable selection

Why Are They Strong Baselines?

- HAR-X (Lasso) is the strongest classical baseline in our study, often ranking #2 or #3 overall
- Daily, weekly, monthly RV lags already encode the key persistence structure of volatility
- Linear models with regularization are much less prone to overfitting than deep networks on small datasets
- Many quant desks use HAR variants as their production volatility forecast, so beating them is a meaningful bar

HAR Model: Fulvio Corsi. 2009. A simple approximate long-memory model of realized volatility. Journal of Financial Econometrics, 7(2), 174 to 196.
HAR-X Framework (exogenous variable extensions): Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. 2023. A machine learning approach to volatility forecasting. Journal of Financial Econometrics, 21(5), 1680 to 1727.
Adam Clements and Daniel PA Preve. 2021. A practical guide to harnessing the HAR volatility model. Journal of Banking & Finance, 133, 106285.
Harvesting the HAR-X Volatility Model (OLS, Ridge, Lasso variants): Adam Clements, Daniel Preve, and Clarence Tee. 2024. Harvesting the HAR-X Volatility Model. Available at SSRN 4733597.
Heterogeneous Market Hypothesis (theoretical foundation): Torben G Andersen and Tim Bollerslev. 1998. Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. International Economic Review, 39(4), 885 to 905.

FAQ

Can M2VN Be Used in Production?

Q: Is M2VN ready for deployment in a live trading or risk management system?

What Is Production-Ready

- Architecture is fully open-source (GitHub)
- End-of-day inference: no real-time streaming required
- Uses only freely available OHLCV data + public news
- TiMaGPT ensures temporal integrity in any backtest
- Deterministic output: single scalar volatility forecast

Gaps Before Deployment

- No uncertainty quantification (point forecasts only)
- Tested on 7 U.S. large-caps; needs broader validation
- TiMaGPT retraining pipeline must be maintained annually
- News sourcing in production requires a licensed data feed
- No online learning: model is retrained periodically

Structural Validity Self-Assessment

✅ Temporal Sanitation: TiMaGPT + FNSPID alignment ensure no future leakage
✅ Dynamic Universe: No survivorship bias in news feed
➖ Rationale Robustness: N/A (M2VN does not generate text explanations)
❌ Epistemic Calibration: Point forecasts only; no uncertainty quantification
❌ Realistic Constraints: Inference cost and latency not yet benchmarked